

Dynamics of Static Stall

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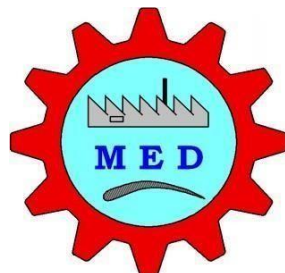
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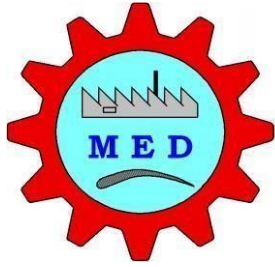
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ABSTRACT

The transition of flow from static to chaotic was studied on a flat plate with a low Reynolds number (Re) regime. This study compares a flat plate of nominal thickness numerically in a flow field with various attack angles (AOA- α).

The flow across a two-dimensional (2D) flat plate with a low Reynolds number and an air velocity (V_∞) of 0.148 m/s is investigated in this work. With varying angle of attack (α)

The results will help understand the stalling phenomenon, how it occurs when the angle of attack (α) varies with flow over the flat plate, and the changes in the flow behaviour associated with it.

Distinct repeating behaviors such as periodicities and irregular cyclicities are natural processes. After some divergence, the recurrence of states is a fundamental property of dynamical systems and is typical for nonlinear or chaotic systems. Recurrence quantification analysis aids in understanding these distinct behaviours.

The frequency and the energy in the leading edge and trailing edge vortex are summarized using different time series analysis technique which is Fast Fourier Transform, Continuous Wavelet Transform, and Empirical mode decomposition. The vortex energy and the characteristic behaviour are studied, that is from coherent to turbulent are distinguish

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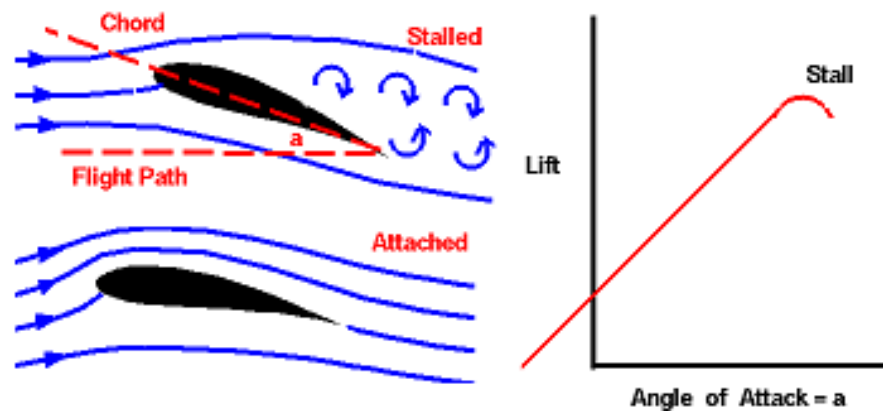
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1. Introduction

A flat plate of negligible thickness is considered for this study. The chord line of this flat plate connects its leading, and trailing edges, the angle between this chord line and the relative wind speed is known as the angle of attack (AOA- α). As the air flows around the flat plate, it exerts a force on it, generating a lift direction perpendicular to the incoming airflow. Lift is accompanied by a drag force parallel to the flow direction, represented by the coefficient of drag (C_d). This lift is a function of the AOA and the flow along with the Reynolds Number. It is described by a dimensionless coefficient known as the coefficient of lift (C_l) that relates the lift generated by a body to the air density around the body and the air velocity. The coefficient of lift (C_l) of the flat plate is proportional to the angle of attack, and this proportionality remains up to a critical point, also known as the 'Stall Point'. The critical point is the angle of attack at which the maximum lift occurs. An increase in the AOA beyond this point substantially diminishes the lift (P.W. Carpenter, 2003).

The stall is a condition such that if the angle of attack increases beyond a certain point, then lift begins to decrease, increasing the drag associated with it. Flow separation starts at the trailing edge as the angle of attack increases and moves towards the leading edge. Separation begins at smaller angles of attack near the trailing edge of the flat plate, while the rest of the flow across the flat plate stays attached. As the angle of attack rises, the separated zones on the top of the flat plate grow in size as the flow separation progresses, impeding the flat plate's capacity to generate lift. This increases the static pressure in the direction of flow, creating an adverse pressure gradient. The adverse pressure gradient and the inlet velocity create favourable conditions for the flow separation over the flat plate.

At stall angle, the adverse pressure gradient is higher, which causes a decrease in C_l and an increase in C_d .



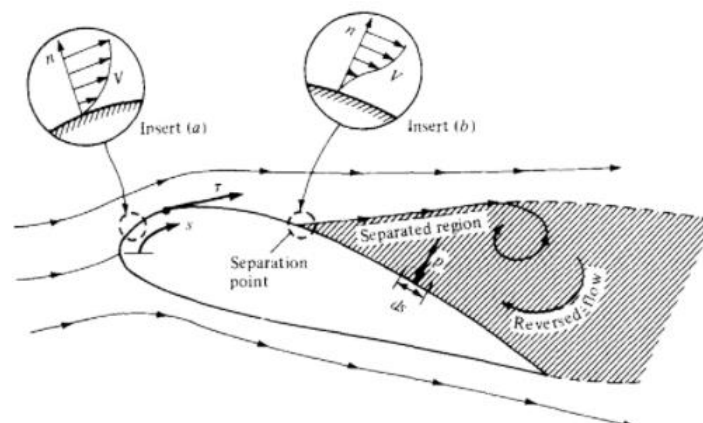
1-1: Relationship between the AOA and lift at critical points, the slightest variation in the angle of attack decreases the lift

The graph depicts a straight line that starts to curve where the value of C_L is highest, reaching its maximum value at the Stalling Point ' α_s ', where α_s is the stalling angle after which the C_L begins to decline (NASA, 2021).

1.1. Phenomenon of stall

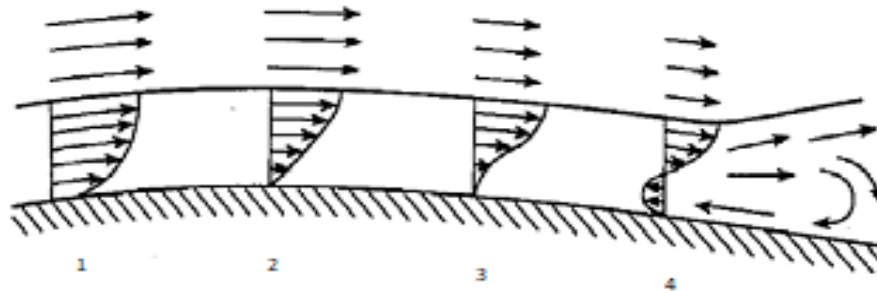
In a flow field, the flow over a body is affected by frictional forces resulting in a reduction in the relative motion over the surface, leading to a no-slip condition just above the body's surface. In the boundary layer, the velocity of the fluid (v) increases as the normal distance (n) over the surface increases till the velocity reaches the maximum flow velocity (v_∞).

Fig 1.2 shows the velocity distribution over a body, and the plot represents the velocity profile (J D Anderson, 2003).



1-2: Phenomenon of Stall

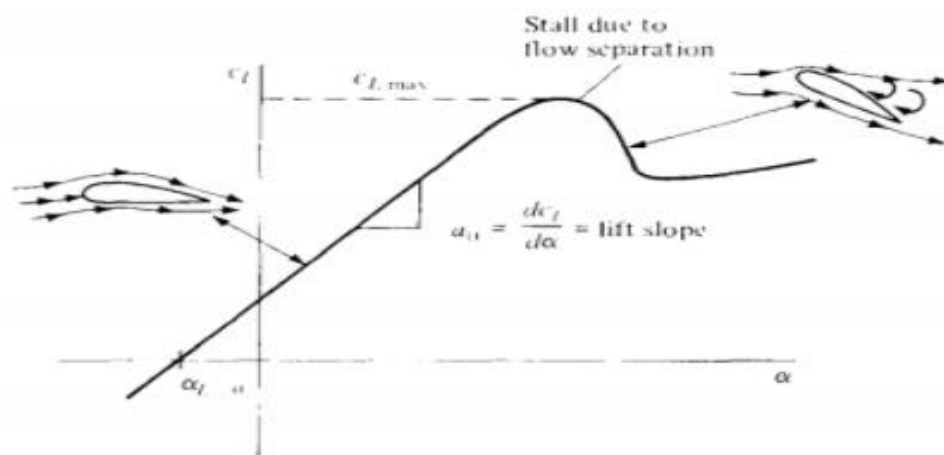
Friction on the surface affects the flow over a body, as shown in Fig 1.3. The initial velocity V_1 is reduced to V_2 and V_3 at points 2 and 3, respectively. An adverse pressure gradient is caused by an increase in the pressure distribution over the body. As the velocity drops below V_3 , the flow direction reverses due to the adverse pressure gradient creating a backward flow with a velocity V_4 (Research Gate , 2014).



1-3: Velocity distribution due to the effect of pressure gradient

The reverse flow increases with the pressure gradient leading to a separation in the fluid flow over the body's surface, creating recirculation wakes downstream.

The flow separation increases upstream as the AOA increases, and as it reaches a critical angle α_s , the wake gets larger. The pressure drag increases the flow separation; a reduction in C_l occurs with a considerable increase in the pressure drag. This phenomenon of an increase of drag with a corresponding decrease in lift is stalling (J D Anderson, 2003).

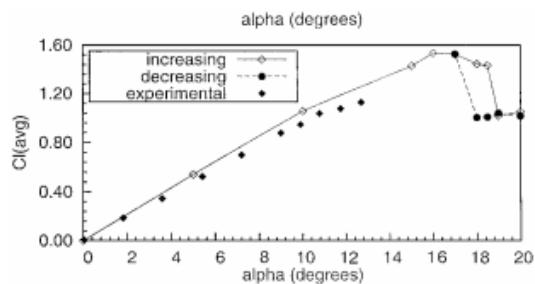


1-4 the stalling in detail

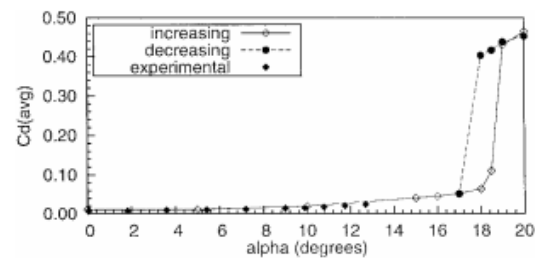
1.1.1. Static stall

As the AOA reaches the critical point, flow separation happens on the body's surface, resulting in a drastic loss in the lift with considerably high drag.

There are two nonlinear dynamics here, jumping and hysteresis. Jumping is the quick change in the response of a system to a higher or lower value with the slightest change in the system parameters (Yan Liu, 2012). Hysteresis is the phenomenon where the value lags behind the change causing it.



1-5 Hysteresis of C_l in static stall



1-6 Hysteresis of C_d in static stall

In the lift coefficient versus AOA (α) plot, at the stall angle, the lift coefficient falls, indicating jumping, and hysteresis occurs from a nonlinear dynamic point of view (Yan Liu, 2012), seen in above Fig (S Mittal, 1999).

Since jumping and hysteresis are the main behaviours of the static stall, the bi-stable condition is seen when these two phenomena co-exist. Hence when a bi-stable condition occurs, it is most likely related to saddle-node bifurcation. Thus the phenomenon of the static stall is related to saddle-node bifurcation (Yan Liu, 2012).

1.1.2. Dynamic stall

Dynamic stall is observed over a body oscillating sinusoidally. The Leading Edge Vortices (LEVs) wind up to form wakes due increase in the shear layer; These LEVs become more unstable and convect away from the body at a higher rate, resulting in reducing the lift (Croskey, 1981).

1.2. *Flow simulation equation*

The behaviour of the flow is complex and is defined using the Navier-Stokes (NS) equations, which have not been concluded effectively yet. Flow is a descriptive variable such as pressure and velocity, which are time and space functions. Thus, it results from a numerical solution of the NS equation; this numerical solution is Direct Numerical Simulation (DNS)(Mahesh, 1998).

The Governing equation for an unsteady 2 Dimension incompressible viscous flow is a sum-up of space, mass, and momentum conservation laws. These equations are known as Navier-Stokes's equation.

Space Conservation Law:

$$\frac{\partial}{\partial t}(\forall) - \frac{d}{dx_j}(\forall u_j^g) = 0$$

Mass Conservation Law:

$$\frac{\partial}{\partial x_j}(\forall u_j) = 0$$

Momentum Conservation Law:

$$\frac{\partial}{\partial t}(u_i) + \frac{\partial}{\partial x_j}([u_j - u_j^g]u_i) = -\frac{\partial}{\partial x_i}(p) + \frac{\partial}{\partial x_j}\left(\left(\frac{1}{Re}\right)\frac{\partial u_i}{\partial x_j}\right)$$

Where,

$\forall$ =Cell volume

u_j^g =Grid Cartesian velocity

u_i =Cartesian velocity vector

p =Pressure

The DNS model resolves the spatial discretization on a temporary scale. The spatial discretization is resolved by the Kolmogorov length scale (η), which is the smallest scale needed to be solved(Wikipedia , 2020).

$$\eta = \left(\frac{v^3}{\epsilon}\right)^{1/4} \quad \dots 1$$

Where v is the kinematic viscosity and is the rate of kinetic energy.

$$\epsilon = u^3/L \quad \dots 2$$

Where u is the root mean square of the velocity, L is the integral scale.

By substituting 2 in 1

$$\eta = \left(\frac{v^3 * L}{u^3}\right)^{1/4} \quad \dots 3$$

For ‘N’ number of mesh points and the increment of h, resolution requirement is satisfied so that integral scale lies in the domain.

$$N * h > L$$

For the Kolmogorov scale to be resolved for the above,

$$h \leq \eta$$

Therefore,

$$N > L/h \quad \dots 4$$

From 3 and 4

$$N > \frac{L}{\left(\frac{v^3 * L}{u^3}\right)^{1/4}}$$

$$N > Re^{3/4}$$

Therefore

$$\eta \sim \frac{L}{Re^{3/4}}$$

Taking L as characteristic length $10 * 10^{-2}$ and Re as 1000

$$\eta \sim \frac{10 * 10^{-2}}{1000^{3/4}}$$

$$\eta = 0.56mm$$

For the simulation, minimum Kolmogorov length scale will be 0.56mm.

To get the minimum time steps

$$C = \frac{u' \Delta t}{h} < 1$$

Where C is the courant number

From which

$$\Delta t = \frac{h}{u'}$$

For $h=0.56 \times 10^{-3}$ m and $u'=0.148$ m/s above equation will be

$$\Delta t = \frac{0.56 * 10^{-3}}{0.148}$$

$$\Delta t = 0.003s$$

In conclusion it can be seen that the minimum length scale of the mesh should be of 0.56mm with the time step of 0.003s according to Kolmogorov and Courant number

2. Literature Review

2.1. *Simulation*

(Nakayama at, 1993):

The numerical study around flat plate at low Reynolds number is carried out. At $Re=200$ the separation bubbles are found symmetrically on both sides of the plate. The Strouhal Number increases linearly with the depth to height (d/h) ratio of plate. At $Re=400$ there is no irregularity found in the vortex pattern and the Strouhal Number increased stepwise with the d/h ratio.

(Vanka, 1995):

Simulation of the unsteady separated flow past a normal flat plate with varying Reynolds number $80 < Re < 1000$, Drag coefficient differed from Re 100 to 1000. For $Re=100$ the vortex patterns travel parallel to the freestream, as Reynolds Number

increases to 500 and 1000, the complex vortex patterns are observed. The time means wake length and drag increases with increase in the Reynolds Number.

(Saxena, 2000):

Study prediction of hysteresis associated with static stall of an airfoil NACA series 0015 with Reynolds number 100 is done. Stalling occurs at AoA 17° for the given airfoil, hysteresis in the aerodynamic characteristics is observed between the AoA 17° and 19° . The results of the simulation of the previous AoA are considered as the boundary conditions for the current AoA when the AoA is increasing. This technique is reversed when the AoA is decreasing. For increasing AoA the flow was almost attached with high lift and low drag observed. Higher unsteadiness and reduction in lift and higher drag was observed during the decreasing order of AoA.

(D.boyd, 2004):

The Flat plate aerodynamic at very low Reynolds number varying from $1 < Re < 200$ were the area of the study made. No separation observed in the fluid flowing across a flat plate even with an increase in the AoA up to 50° and a very low Reynolds Number (Re) of 1.375. Thus the aerodynamic characteristics of the flat plate were found to be very poor at low Reynolds number, and the maximum lift was at AoA 20° .

(Yan liu et.al, 2011):

Numerical bifurcation analysis of static stall of aerofoil with NACA0012 with Reynolds number 1000 are simulated with varying angle of attack. Study proves that Static Stall is a saddle-node bifurcation, with jumping and hysteresis as the primary characteristics.

(Karthik Menon, 2019):

Aerodynamic characteristic study of different aerofoil with different Reynolds number is studied.

(D Durante, 2020):

Bifurcation and chaos transition of the flow over an aerofoil at low Reynolds number of 1000 and varying angle of attack on NACA00012 is studied. Flow over NACA airfoil with varying angle of attack with low Reynolds number. The different modes of flow were identified, and the flow transition from periodicity to chaos from AoA 26.5° to 27.1° , and inverse transition from AoA 27.1° to 28.5° also found.

(Liu, 2021):

Flow over a flat plate with Reynolds number 1000 and varying angle attack from 0° - 50° in a regime. The results were compared with the NACA aerofoil. It was found that there is a sudden increase in the lift coefficient after an angle of 32.5° compared results were the same profile.

2.2. *Data Driven Techniques*

RQA

(Fraser A M, 1986):

Independent Coordinates for Strange Attractors from Mutual Information. A mutual information calculation algorithm is described. The first minima are found to be an effective criterion for selecting a time delay for reconstructing time series data. This criterion as compared to the autocorrelation function produces better results.

(J. P. Eckmann, 1987):

Study over Recurrence Plots of Dynamical Systems. Recurrence Plots are proposed as a distinct graphical tool for measuring the time constancy of dynamical systems (RPs).

(Mathew B. Kennel, 1992):

Determining embedding dimension for phase-space reconstruction using a geometrical construction, the behaviour of near neighbours under changes in the embedding dimension from $d \rightarrow d+1$ is examined in order to determine an acceptable minimum embedding dimension. This information is utilized in the analysis of the observed time series.

(Joseph S. Iwanski, 1998):

Recurrence plots of experimental data: To embed or not to embed? is addressed. Recurrence plot is suggested as a visualisation tool for interpreting experimental data, with emphasis on the significance of proper embedding and its importance in the selection of dimension.

(Joseph P Zbilut and Charles L Webber, 1992)

Recurrence Quantification Analysis: Introduction And Historical Context are the study area. Traditional analytic methods can make it difficult to analyse chaotic systems. Hence it is suggested that recurrence plots be used for determining recurrence quantification based on distance matrices of the embedded series. Also showcases the plot's ability to visualise data.

Time series analysis

(Norden E. Huang, 1998):

EMD is proposed for the decomposition of non-linear and non-stationary data into different levels in order to get intrinsic mode function with shifting techniques, for the analysis of local frequency.

(Tzu-Hsun Hung, 2012):

The time series data are decomposed into several IMFs using Empirical Mode Decomposition (EMD) algorithm. Hilbert transformation algorithm is utilized to acquire the instantaneous frequency for every IMF.

(Nicolas Mazellier, 2013):

Discussion about the EMD in approximate discrimination between the coherent and turbulent contribution in the signal

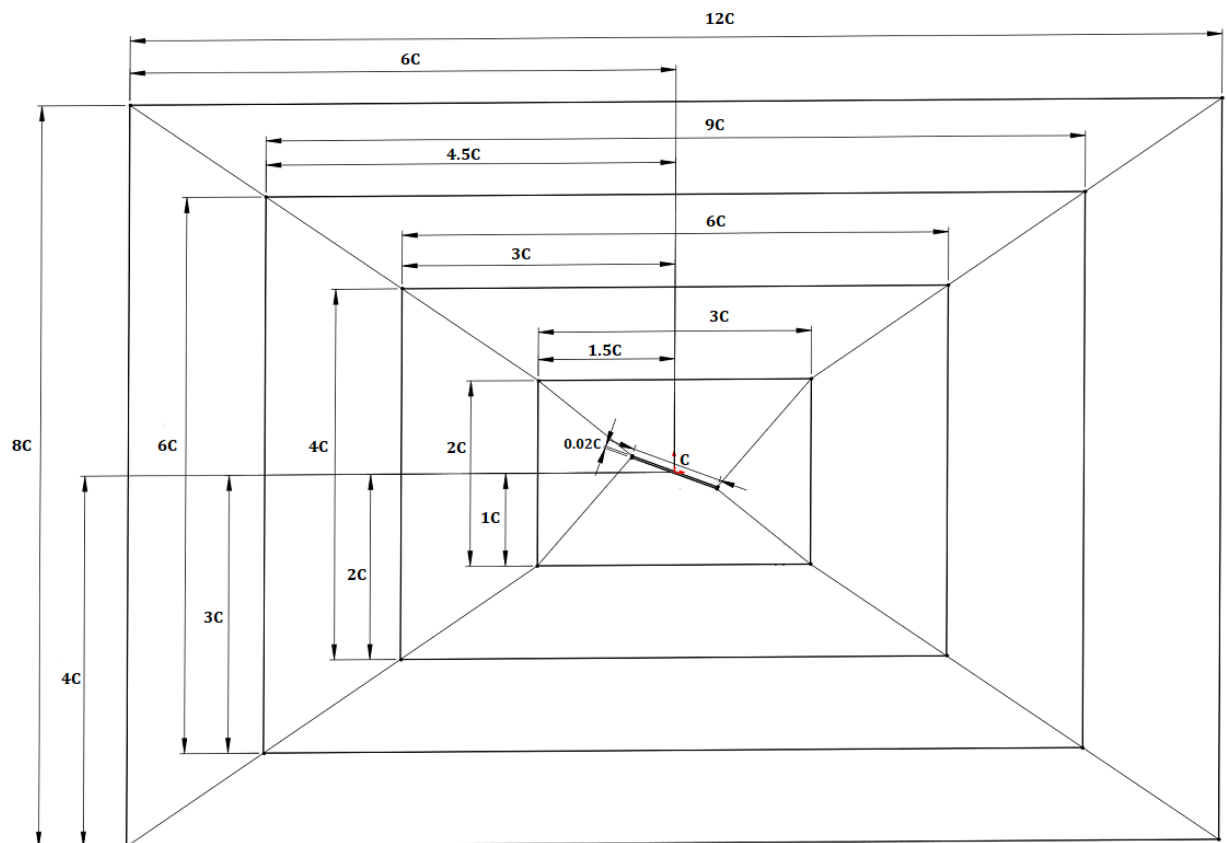
(Subash Chandran KS, 2016):

Several different techniques have been used to optimize the time frequency, improvement in the spectral and temporal resolution to compare with other methods.

3. Grid Generation and Simulation Details

The two dimensional (2D) computational rectangular domain with $12C \times 8C$ in size is created. The upstream and the downstream length of $5.5C$ and $6.5C$ was selected respectively. upper and bottom boundaries of the domain have four times the chord lengths. These dimensions are ascertained to not have any effect on the flow generated.

The geometry of the domain is smaller than the background mesh. The geometry of the domain is shown in the figure.



3-1 Geometry model

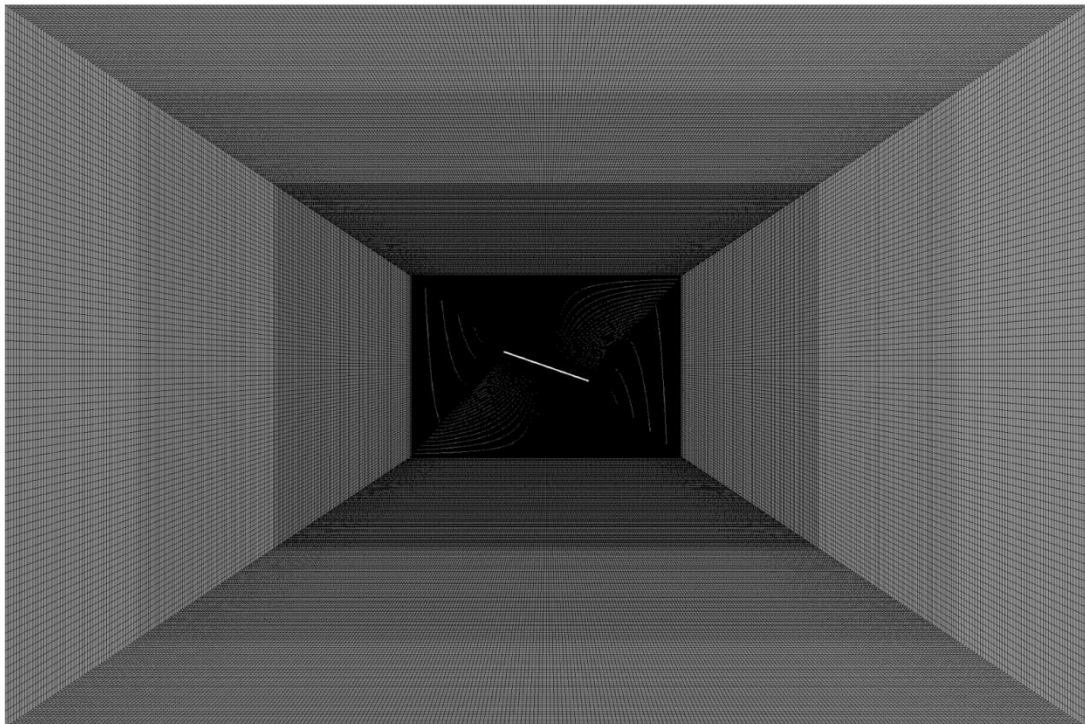
The flat plate model shown in Fig. is done using ANSYS Workbench. The rectangular plate having dimensions of 10×0.2 cm is placed in the centre of two-dimensional computational domain. A face split is done to project the selected faces into multiple separate faces, which enables them to select each face separately. This separation helps in meshing the faces with the required quality of mesh based on the accuracy of solution required.

The domain is equally divided increasingly from the inner most rectangles to the overset. The inner rectangle of dimension $3C \times 2C$ is increased till $8C \times 12C$ with a linear increasing pattern. This provides the better control over the mesh size

The position of the plate is such that the downstream region is same as the upstream region. This is particularly done as the vortices generated around the plate move downstream and is required to have sufficient room to accommodate all the vortices generated, the plate rest at the centre of the domain. The angles considered for this study are 0° to 32.5° .

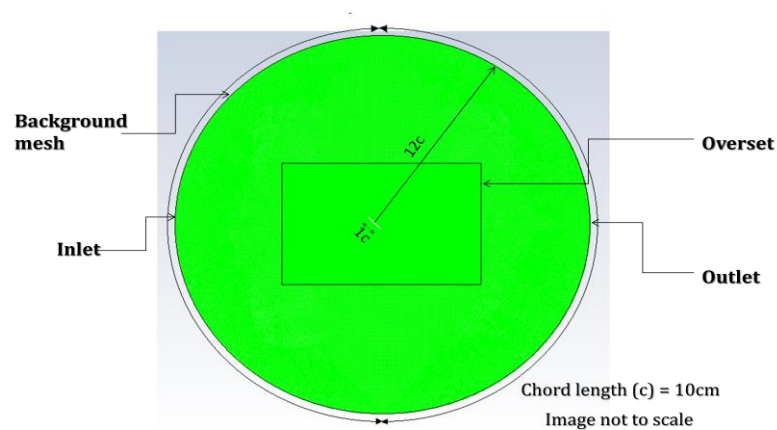
The meshing of the domain is achieved using the sizing function in ANSYS Workbench. The sizing depends on the coarseness, or the fineness of the mesh required at that particular region. This is achieved by incorporating a combination of details such as number of divisions, bias type, and bias factor.

The cells in the outer parts need to be uniformly spaced where there are no vortices. The fineness of the mesh increases closer to the plate with the number of cell divisions highest around the plate. This provides better accuracy as most of the vortex formation is around the plate before it gradually moves downstream.



3-2 Mesh Details

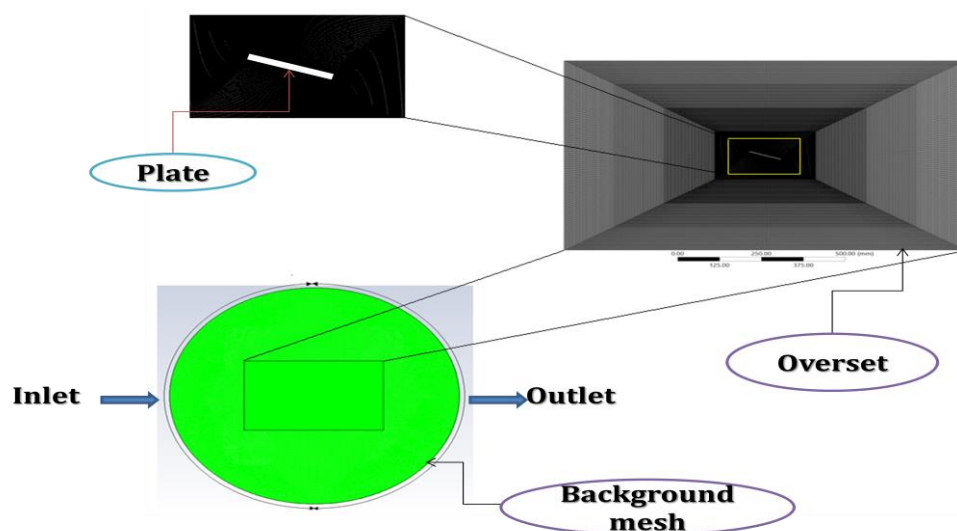
Background mesh and overset:



3-3 Background mesh with overset

A 2D circular background mesh with radius of 12-time chord length is created using ANSYS Workbench. The circular geometry was constructed with 2 semicircles. First half circle (left) is named as the inlet and the second half circle (right) is named as the outlet. The background mesh comprised of 2.02 lakhs nodes. The overset domain used to append on this background mesh with the help of the append tool in fluent. The reason to have this background mesh is to get the clear insight of far flow field data to investigate the flow field accurately.

Simulations and boundary conditions:



3-4 Detail of mesh in Fluent

The Ansys fluent was used for simulation. Fluent is used to solve the equations. The parameters selected for obtaining the required solution are double precision, parallel solver with 8 processes, with laminar flow and material selected is air. DNS(direct numerical simulation) model is used because of low Reynolds number and laminar flow.

Fluent is used to solve the equations. The parameters selected for obtaining the required solution with double precision, laminar flow and material selected is air. The boundary conditions are set for inlet as inlet velocity (v) of 0.148 m/s, an exit pressure as pressure outlet, and the plate is considered as stationary wall with no slip condition.

Then reports generated are that of lift force and drag, which help determine the coefficient of lift and drag respectively. This study is conducted under the following conditions: iterations converge up to the order of 0.000001, size of the time step is 0.003s with 100000 number of time steps, with 20 iteration/time steps.

For animation, first go to user define and select custom field function, in that select derivatives and select $(dy\text{-}vel/dx - dx\text{-}vel/dy) * 0.5$ and name it as wz.

Then select velocity >> (vorticity magnitude*vorticity magnitude), derivatives >> (strain rate * strain rate) and complete the equation $((vorticity\ magnitude * vorticity\ magnitude) - (strain\ rate * strain\ rate) * 0.5$ and saved as q

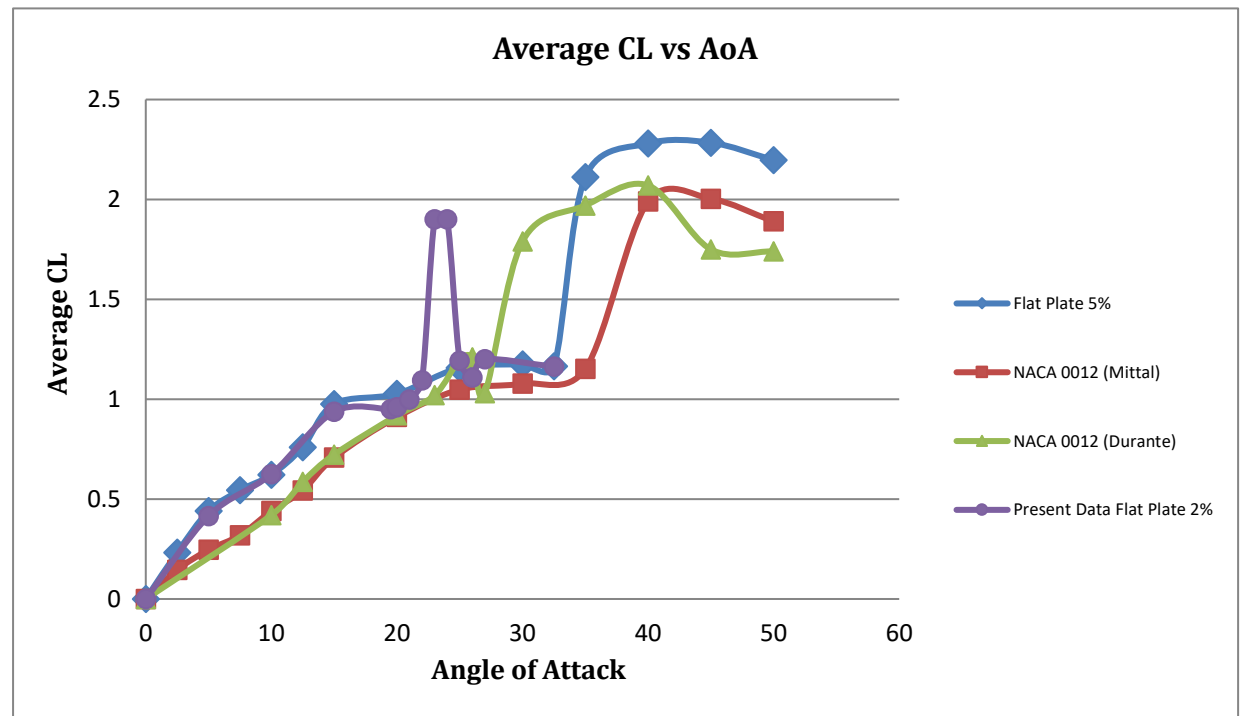
Then pick domain >> surface >> create >> iso clip >> custom feed function >> select q >> interior surface body 1, give minimum value of 0 and maximum value of 50, name it as q50, and click create.

After that, go to graphics. Select contour and wz, then select q50 and plate, set the minimum and maximum values to -0.5 and 0.5, and save/display.

Then go to calculation activities and pick solution animations, then choose contour1 and preview and user active. Then go to animations, pick MPEG, and click write, and the animation will be saved to the specified directory.

4. Results and Discussion

4.1. Validation

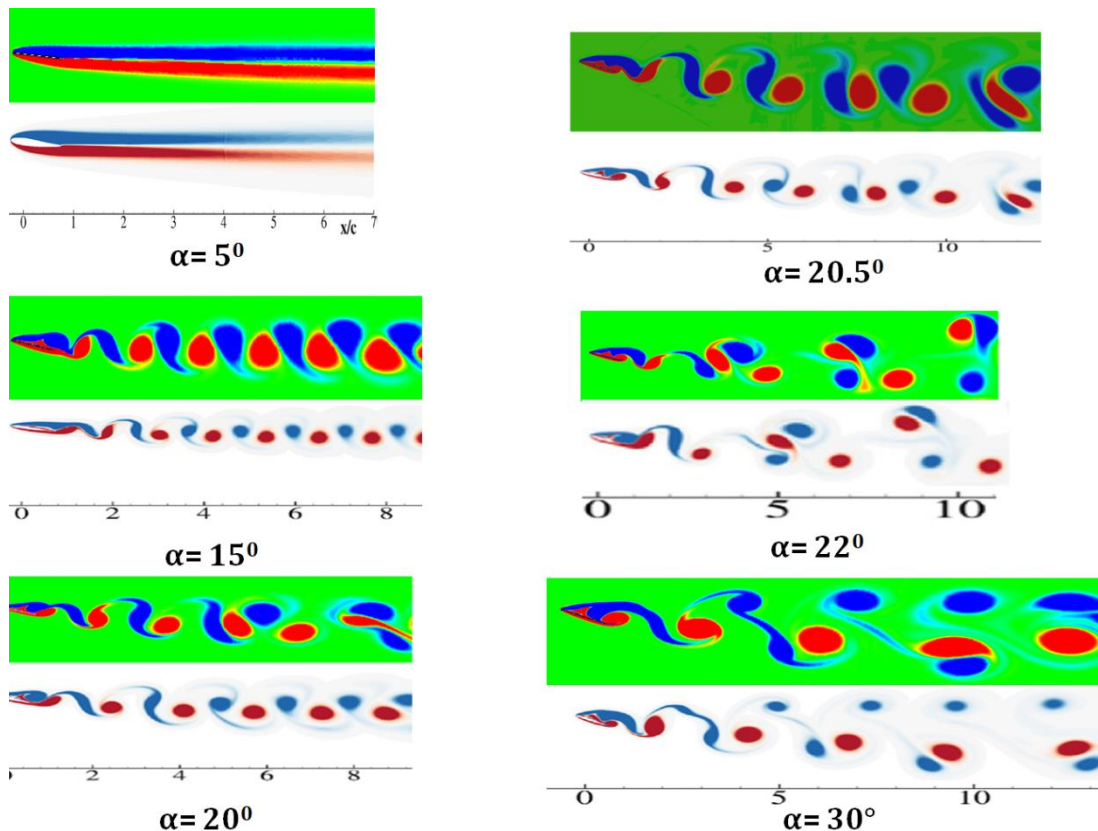


4-1 Cl comparison with Mittal, Durante, and flat plate 5%

Three journal papers were considered to validate our results. The above graph shows the comparison of the results obtained in the present simulation and for Flat Plate with chord length of thickness 5%, NACA 0012 from Mittal and NACA 0012 from Durante. From the comparison it can be observed that the data of present simulation and flat plate 5% are of same trend with marginal change in Cl values for the same AoA and the present simulation shows that there is a sudden increase in Cl value from an AoA of 22° to 23° and the increase is extended till 24° after which a sudden drop is observed.

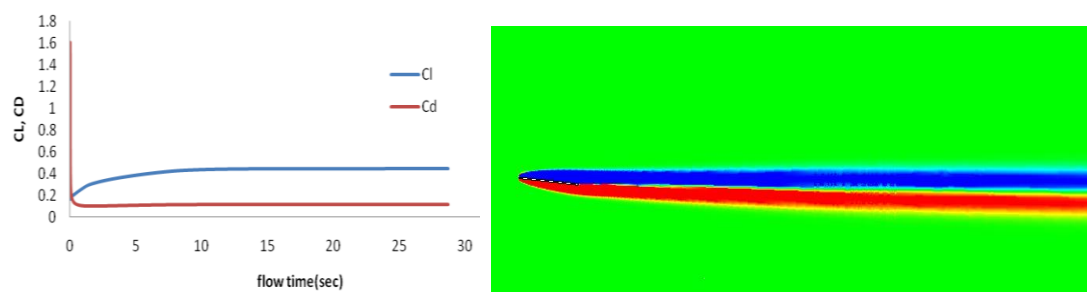
This validates the data obtained and the trend in Cl value is same for all the comparison values. A detailed study can be done between AoA 22° to 25° based on the comparison

4.2. Flow field



4-2 Vortex patterns of the different angle of attack from our study (green colour) compared with the D. Durante (below).

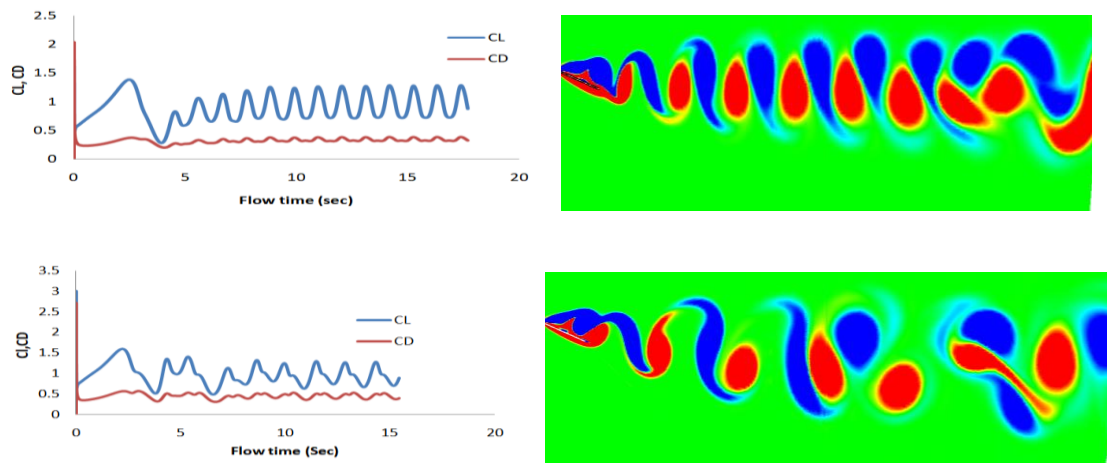
Starting from 0° to 9° there are two stable shear layer of vorticity field is found. The flow around the flat plate found to be attached. Since the angle of attack is small there is no flow separation on the plate, and the wake remains narrow and stable. Both the lift and drag signals are in steady state. Because of the shedding mechanism not yet started the force time signals are constant straight lines.



4-3 Time histories of C_l & C_d at AoA 5° , corresponding vortex pattern

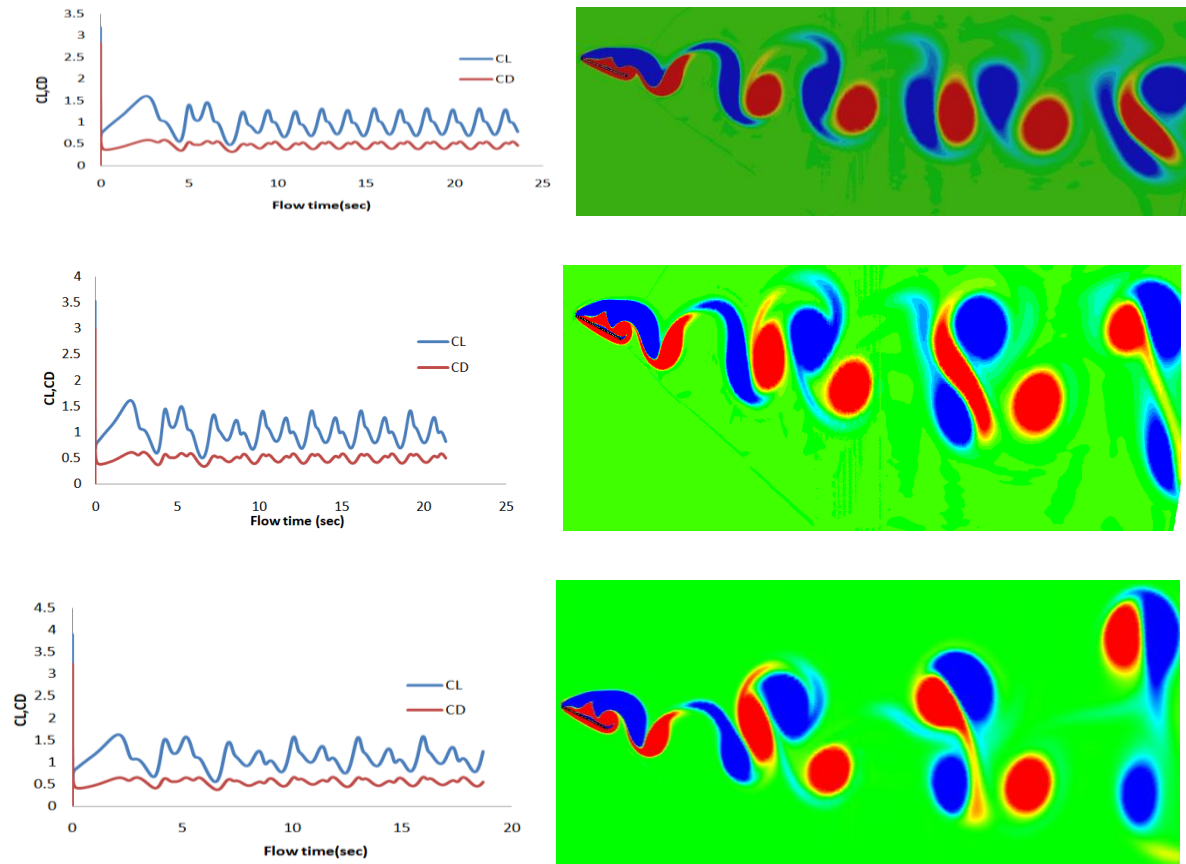
As the angle of attack increased further the periodic Karman vortex shedding is observed from 10° to 20.5° . In this stage the time signal is became slightly unstable they start to oscillate in periodic sinusoidal manner. In this stage the trailing edge vortex TEV start to climb upward by shedding the leading-edge vortex LEV. The two opposite rotating vortex pair are the major vortex in this mode.

As AoA goes to 20° and 20.5° , even the vortex shedding is periodic and alternative there is unstable in the far flow field is observed. The lift signal is characterized by the two local maxima (peaks) in the time series.



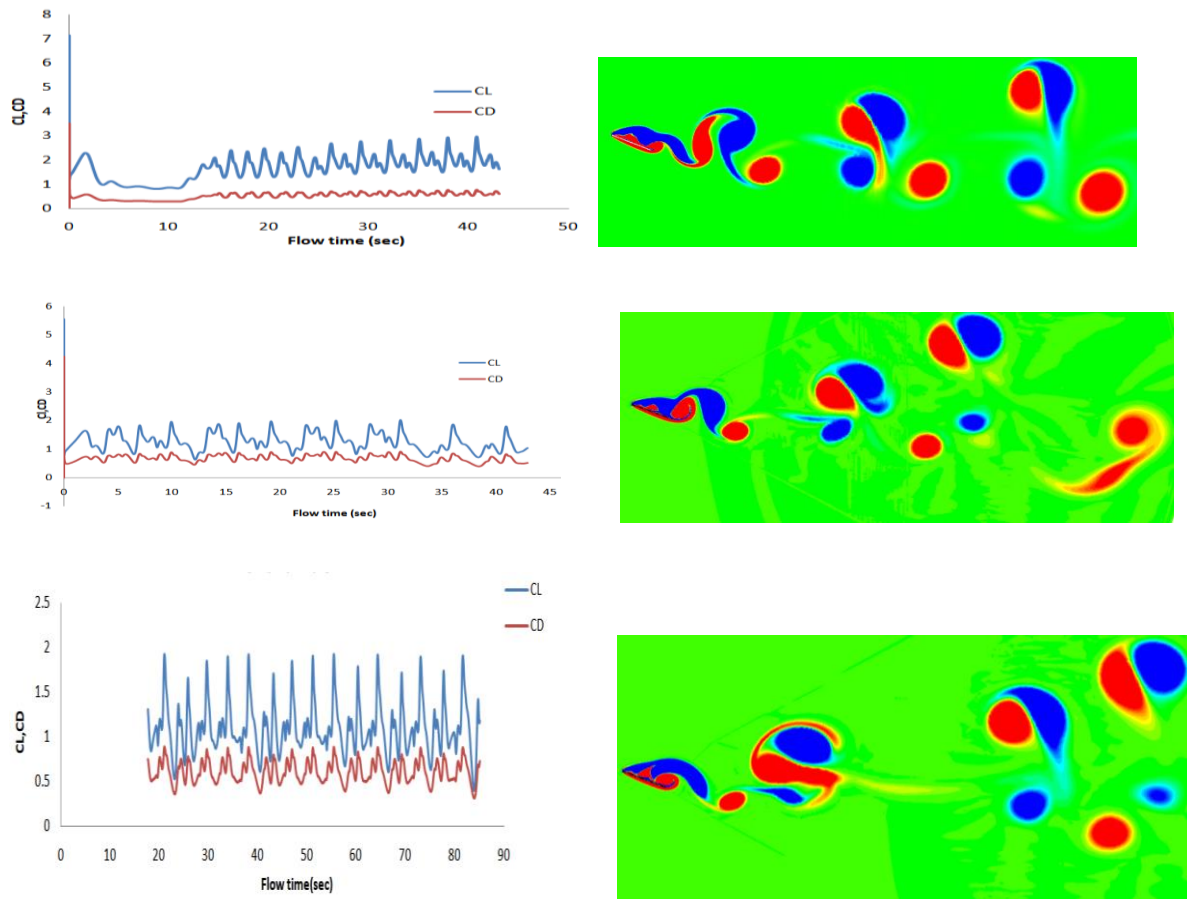
4-4 time histories of c_l & c_d at AoA 15° , and vortex structures (top). Time histories and vortex structures at AoA 20° (bottom)

From 20.5° new pattern of vortex shedding is recognized. The flow field becomes more unstable and the complex vortex patterns started to appear. The two opposite rotating vortex pair (dipole) are formed, and they bent streamwise and started to climb upwards by deviating from the wake centre line. A dipole with the single vortex is the major vortex pattern in this stage. For this corresponding flow field, the lift signals are characterized by the three local maxima (peaks). At the angle of attack of 22° the lift signal found to be more complex, and this is the sign of the flow stated to enter the chaotic regime.



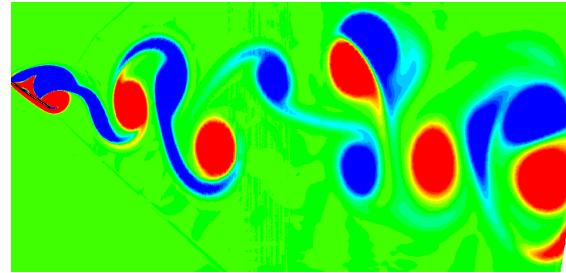
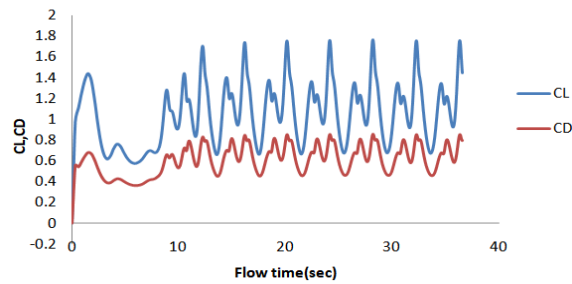
4-5 Time histories of cl & cd at AoA 20.5°, and vortex structures (top). Time histories and vortex structures at AoA 21°(centre). Time histories and vortex structures at AoA 22°(bottom)

Starting from the AoA 23° the flow field becomes more complex, the vortices are unpredictable, and at this point from AoA 22° to 23° there is a sudden jump was found in average lift signal with respect to the AoA , and 23° to 24° there is a linear trend is observed, from 24° to 25° again there is a sudden dip is found. This is because of at this AoA the plate reached its Stall angle. There is a greater fluctuation is found in the lift signal, unlike the previous stages there is no repeating pattern are recognized. This confirms that the flow is reached the chaotic condition.



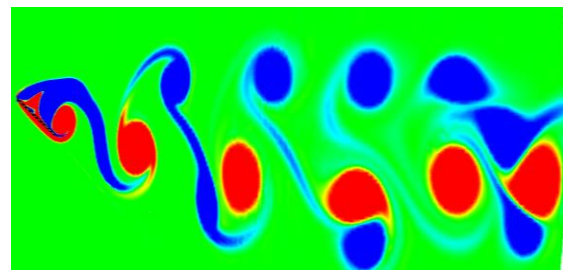
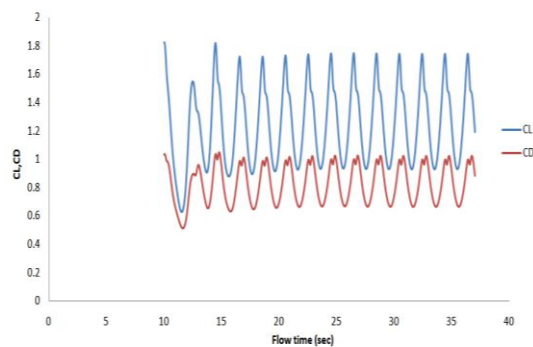
4-6 Time histories of c_l & c_d at AoA 23°, and vortex structures (top). Time histories and vortex structures at AoA 24.7°(centre). Time histories and vortex structures at AoA 25°(bottom)

At the AoA 26° and 27° there is still some unsteadiness is present in the flow field that is clearly visible from the vortices that are formed. At this stage, a vortex dipole with the two single vortex is recognized. The dipole is first formed and that start to deviate away from the wake centre line. At this stage, the lift signal is characterized by the 4 local maxima and 2 local maxima for 26° and 27° respectively. And, the lift signals seem to be periodically repeating; however these repeating patterns are not seen from the previous stages, from this we will say that the flow is started to stable slowly.



4-7 Time histories of c_l & c_d at AoA 26° and vortex patterns (top). Time histories of c_l & c_d at AoA 27°

As further increase in the AoA to 30° , in flow field the vortex dipoles found with a single vortex, where the dipoles are formed and propagate downward by deviating from the wake centre line. Where the upper vortex are the clockwise rotating vortices. The lift signal at this stage is slightly stable and found to be repeat periodically. The lift signal is characterized by the 3 local maxima(peaks).



4-8 Time histories of c_l & c_d at AoA 30° (left), corresponding vortex structures.

5. Recurrence Quantification Analysis

A recurrence is a state where the trajectory of a point in a time series returns to a location it has visited before. Recurrence Quantification Analysis is a method that can be applied in the study of temporal correlation within a time series data and the sequential correlations within the data sets. The fundamental objective of recurrence analysis is to find recurrences or patterns in dynamical systems that recur throughout time to the same or similar states due to data set movements. These changes are documented using RQA. Based on repeatability of time series states it presents the time series data set as a geometric structure. The topology of such geometric structures enables the discovery and analysis of the characteristics of time series' dynamics.

The RQA measures can be an early indicator of the beginning of chaos in a system since it provides a solid indication of the transition from the periodic phase to the chaotic phase with the transition regime being carefully captured.

Steps:

- i. Collections of time series data
- ii. Projecting the time series data into phase space
- iii. Plot recurrences
- iv. Quantify the pattern of recurrences

5.1. Phase space

A phase space is a multidimensional space representing every degree of freedom or parameter of a system as an axis. One and two-dimensional systems are called phase line and phase plane, respectively.

Over time, the evolution of the system's state can be traced as a path in this phase space with corresponding higher dimensions. The phase diagram, as a whole, represents everything that the system can be, and its shape can reveal features of the system that would otherwise be obvious.

5.2. *Takens' theorem*

Time-delay embedding is done in order to reconstruct an attractor, Takens' theorem (1980). Suppose a dynamical system contracts onto an unobservable attractor contained within a lower-dimensional manifold of unknown dimension N , the observed time series, called $x(t)$, lies on this attractor. The time-delay embedding procedure uses $x(t)$ to reconstruct the attractor with lagged coordinates τ , where τ is the length of the lag, and an M number of embedding dimensions.

Takens' theorem states that, "For the right τ and enough dimensions, the embedded dynamics are diffeomorphic or have the same topology as the original phase space dynamics. The measured quantity must be a smooth generic function of at least one state variable and must be uniformly sampled in time."

5.3. *Delay Parameter (τ) and Embedding Dimension (M)*

The unfolding of the one-dimensional time-series into a multidimensional phase-space is done to ensure that reconstructed dimensions of the phase space are relatively orthogonal and deliver non-confounded information about the temporal structure in a time-series. A phase space is constructed by plotting delayed time-series data against the original one. A collection of data-sets is created from a single data-set, each being a delayed version of the original called surrogate data. Two embedding parameters τ and m are chosen for this, τ sets the primary delay and m the number of embedding dimensions.

Takens theorem states that delay embedding accurately generates the state variables, provided τ and m are appropriately chosen. However, the two embedding parameters are usually adjusted to get good results in the subsequent steps. If τ is too small, the constructed signals are very similar, and if it is too large, they have no relationship (Joseph S. Iwanski, 1998). Hence, particular procedures are used to select these parameters.

5.3.1. *Average Mutual Information (AMI)*

This method is used to find the time-delayed coordinates that are maximally independent of each other. The dependence that exists is quantified as mutual information $I(x(t), x(t+\tau))$ between the original time series $x(t)$ & $x(t+\tau)$ shifted by τ

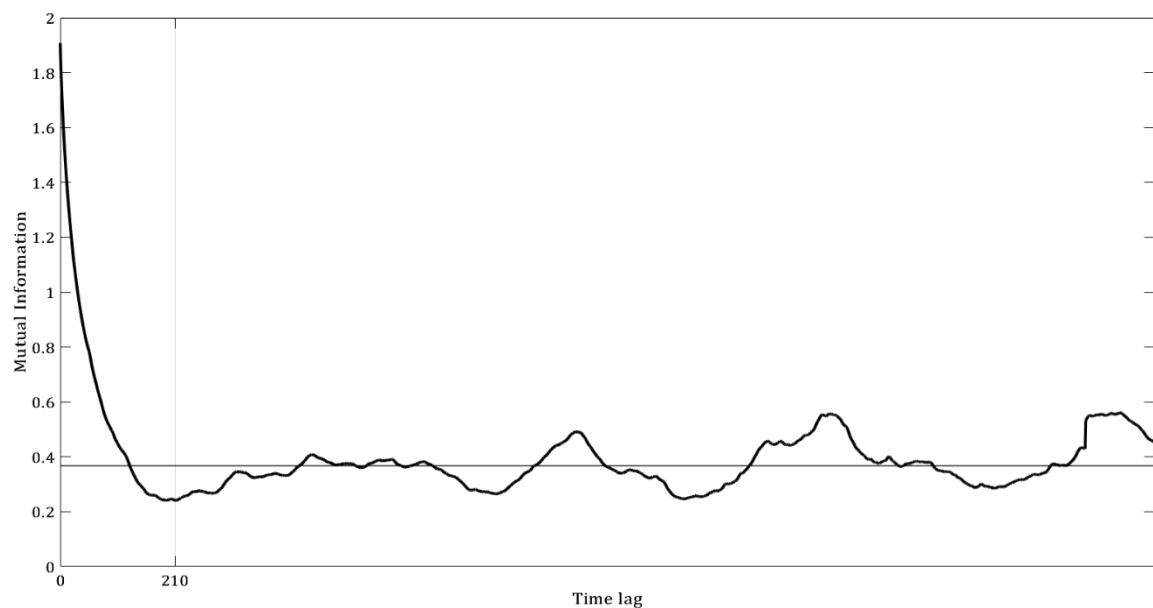
$$I(x(t), x(t + \tau)) = \sum_{ij} P_{ij}(\tau) * \log \left[\frac{P_{ij}(\tau)}{P_i P_j} \right]$$

P_i is the probability that $x(t)$ is in bin i

$P_{ij}(\tau)$ is the probability that $x(t)$ is in bin i and $x(t+\tau)$ is in bin j

To obtain coordinates for time-delayed phase space embedding the first minima position of $I(x(t), x(t+\tau))$ was proposed, meaning the first coordinate of the phase space embedding signal $y_1(t) = x(t)$ is considered since it is maximally independent of the second coordinate $y_2(t) = x(t+\tau)$

This creates enough independence between two successively lagged values, and ensures that it isn't short lagged with unnecessary information or overly lagged with too little information to construct the attractor diagram (Fraser A M, 1986).



5-1 Graph for Average Mutual Information

The above figure shows the mutual information graph in which a delay factor of 210 is considered.

5.3.2. *False Nearest Neighbour*

The embedding dimension M is chosen using the ‘false nearest neighbours’ method. This method measures the percentage of points that are close in a given dimension that remains so in the next higher dimension too. If the embedding dimension is capable of containing the reconstructed attractor, then the percentage of false nearest neighbours drops to zero.

Using the original time series $x(t)$ we can obtain $y(t)$ time series of M -dimensional points using the replace time-delayed copies of the original as follows:

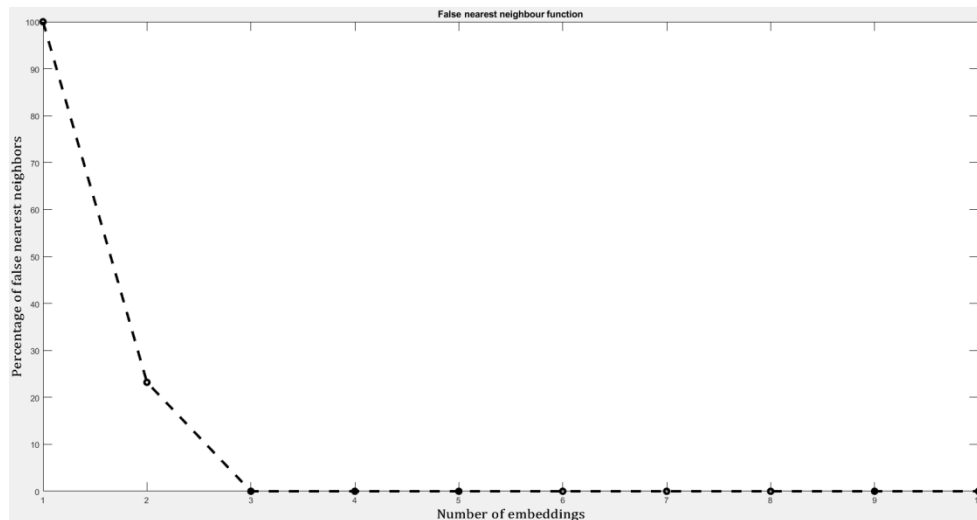
$$Y(t) = (x(t), x(t+\tau), \dots, x(t+(M-1)\tau))$$

Embed dimension M , by examining the distance between neighbouring points in the phase space as the original time series is progressively embedded into higher dimensions.

Suppose two data set points in a one-dimensional time series are close together i.e. neighbours. Their difference in magnitude provides the distance between those neighbours. When the time series is embedded once into two dimensions using a time delay τ , the coordinates of those data points can examine whether the distance between them has changed appreciably. If embedding changes the distance between the neighbours appreciably, they are dubbed false neighbours, indicating that the embedding must continue.

If embedding leaves the shape of the attractor unchanged and the distance does not change appreciably, then the current embedding is sufficient, and the coordinates are dubbed true neighbours.

This condition is achieved by consecutively increasing the embedding dimensions D to a point where the FNN drops to zero and subsequent embeddings do not change the value of FNN or if it starts to increase again (Mathew B. Kennel, 1992).

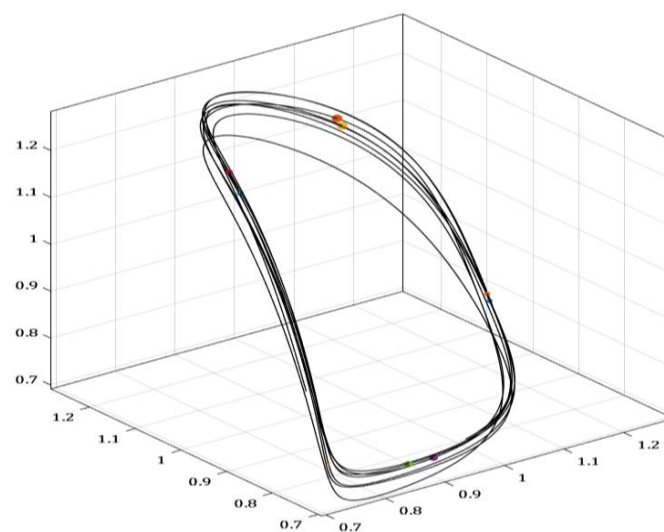


5-2 False Nearest Neighbour graph indicating Number of embeddings.

The above graph of Percentage of the Nearest Neighbour vs Number of Embeddings shows the number of dimensions embedded for the case taken. 3 number of dimensions is considered as it provides the suitable number of embeddings to reconstruct the given time series in a phase space.

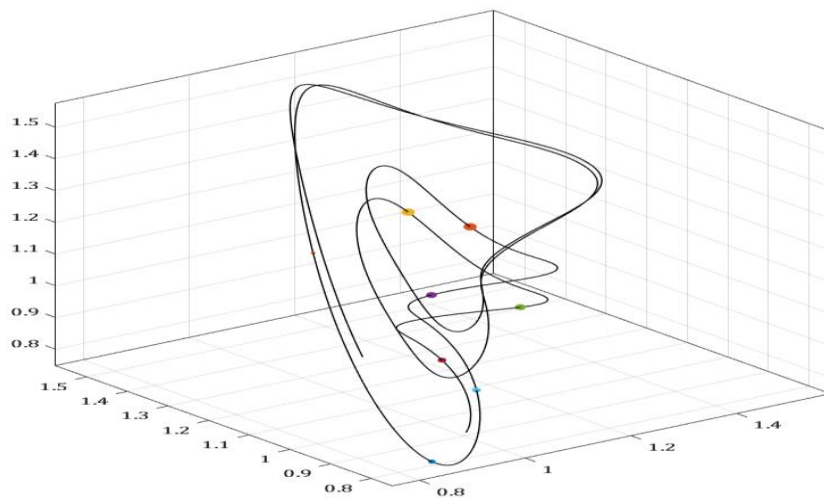
5.4. Attractor diagram

An attractor diagram represents the dynamics of a system. A joint probability matrix P_{ij} reduces the n-dimensional initial problem by compressing the dynamical information into 3-dimensions. This process offers an ideal representation of the problem while maintaining quantitative and qualitative information.



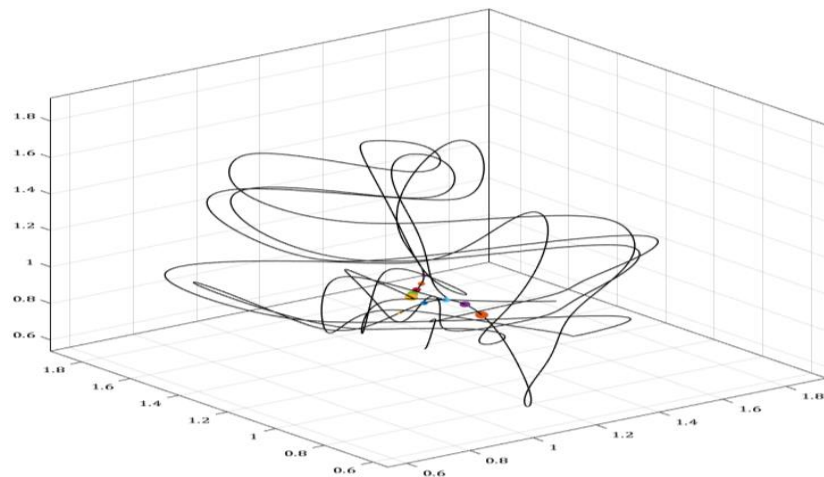
5-3 Attractor Diagram for Angle of Attack 10°

The attractor diagram in the above figure is for an Angle of Attack of 10° . The attractor diagram is smooth indicating that the time series is periodic.



5-4 Attractor Diagram for Angle of Attack 22°

The attractor diagram in the above figure is for an Angle of Attack of 22° . The attractor diagram is mostly smooth while there are some elements of chaotic behaviour. This indicates that there is a transition from periodic to chaotic behaviour happening.



5-5 Attractor Diagram for Angle of Attack 25°

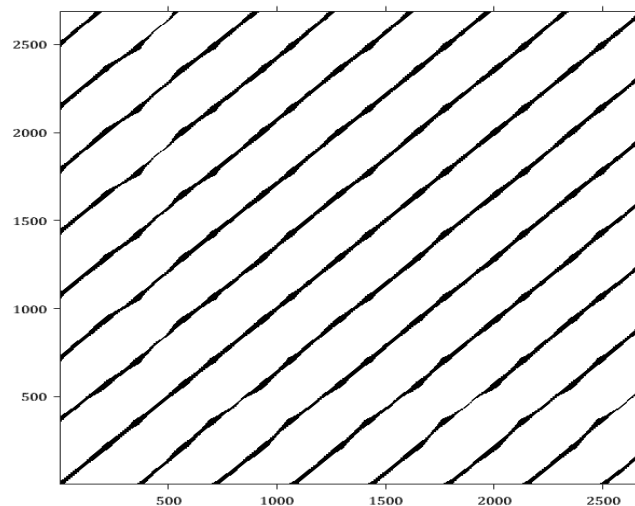
The attractor diagram in the above figure is for an Angle of Attack of 25° . The attractor diagram is completely irregular, without any noticeable patterns visible. This indicates that the transition is complete to chaotic behaviour. Thus the coefficient of lift is decreasing while the coefficient of drag is increasing.

5.5. Recurrence Plots

RPs visualizes the trajectories in phase space and show the distance between states of the system (J. P. Eckmann, 1987). For each moment 'i' in time the times at which a phase space trajectory visits roughly the same area at time 'j' is plotted. Principally it is a graph of $\vec{x}(i) \approx \vec{x}(j)$ where x is the phase space trajectory. The recurrence plot shows a series of times where the trajectory is at the same location.

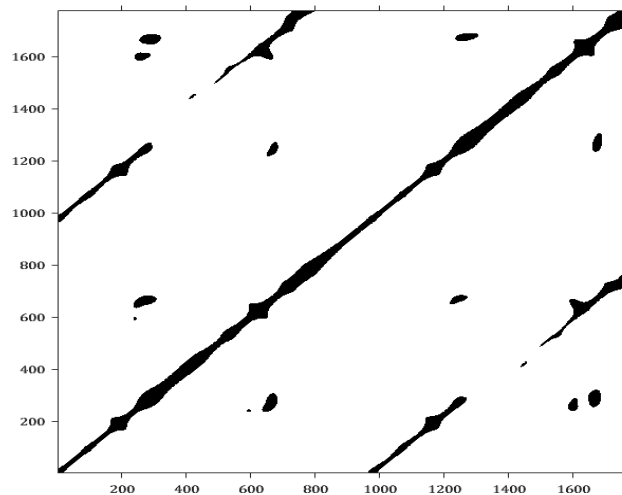
Recurrence quantification analysis uses small-scale structures in RPs; this quantification enables its description and investigates system transitions quantitatively (Joseph P Zbilut and Charles L Webber, 1992) (Marwan, 2002). Recurrence plots have the advantage of providing relevant information even for brief and non-stationary data, where other approaches fail.

Its initial purpose is to analyze higher-dimensional phase space trajectories visually. The perspective on RPs provides information about how these trajectories have evolved through time. This method can be applied to both short as well as non stationary data.



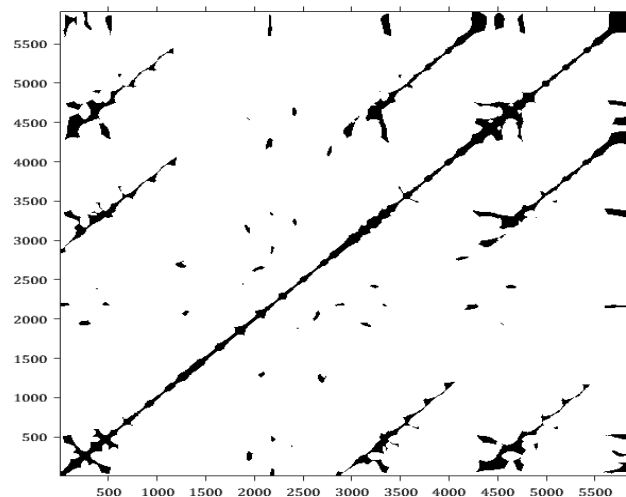
5-6 Recurrence Plot for Angle of Attack 10°

The recurrence plot for the Angle of Attack 10° shows uniformity throughout which suggests a repetitive pattern in the data set.



5-7 Recurrence Plot for Angle of Attack 22°

The recurrence plot for the Angle of Attack 22°, shows partially chaotic behaviour with some elements of periodicity suggesting a transition happening.



5-8 Recurrence Plot for Angle of Attack 25°

The recurrence plot for the Angle of Attack 25° shows complete chaos with almost no patterns indicative that there are very few recurrences happening.

A transition is observed in all the above recurrence plots from periodic behaviour to chaotic behaviour. Thus recurrence plots are a tremendous tool to identify trends and behaviours.

5.6. Recurrence Parameters

Angle	Dimension	Delay	Recurrence Rate (%)	Determinism (%)
15°	3	86	10.22	99.40
22°	3	90	6.62	97.86
25°	3	210	5.4	93.58

The recurrence parameters table exhibits decreasing rate of determinism, which is a key indicating factor of the recurrence in a given data set. Notice for Angle of Attack of 15° the determinism rate is high as the AoA is increased the determinism rate comes down. Thus, it can be inferred from this that the Coefficient of Lift is decreasing and fewer recurrences happening in the given data set, hence chaos has developed.

6. Time Series Analysis

There are two kind of vortex that are associated with in flat plate which are leading edge vortex (LEV) and trailing edge vortex (TEV).

Leading edge vortex are coherent in nature, will be of low frequency in lift, flow dominant vortex which will have low Strouhal number. Whereas the trailing edge vortex are turbulent in nature and will be of high frequency contribution in lift. Trailing edge vortex are because of flow separation, so it will be viscous dominant overflow vortex which is also known as Karman vortex shedding. It is very important to study that the frequency and energy contribution of vortex, so time series analysis is used in order to get inference from the data's obtained.

Time series are the signals that change over time and contain information. Linear, nonlinear, stationary, and non-stationary are some of the types of time-series signals.

Stationary signals are those where the mean and variance remain the same over time. In comparison, non-stationary signals change their mean and variance with time, which allows a detailed analysis of it. Linear signals are the one which correlates the data for an extended period. Nonlinear signals are those where the data diverts from the

correlation increasingly with time. The data set obtained a mean and variance changing over a period, indicating that the data are non-stationary in behaviour. The prediction of the data is more complex, which indicates a nonlinear signal.

Signals and their attributes obtain from simulation data change over time. Such time-series signals are studied using Fast Fourier transform (FFT) to analyse their frequency and the energy stored in that signal. A further detailed study is carried out through Empirical mode decomposition for a comprehensive understanding.

6. 1. *Fourier Transform*

Any finite energy signal is decomposed using the Fourier transform as a linear combination of sin and cosine functions at various frequencies, which are the basic functions.

The above equation is the Fourier transform, where ω represents the angular velocity. The Fourier Transform converts the data into sin and cosine terms by the complex equation where the energy distribution of the signal is identified. The Fourier transform data can be stacked in an n by n matrix, where n is the total number of available data sets.

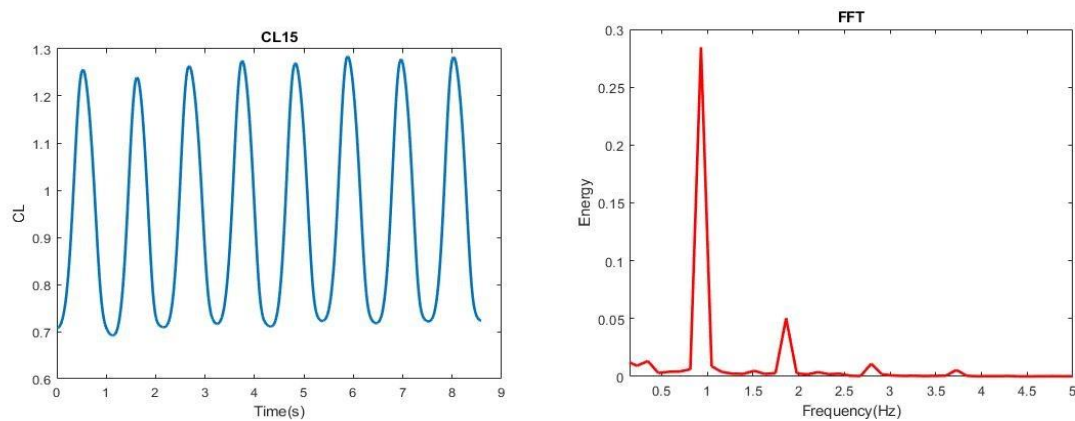
6. 2. *Fast Fourier Transform*

The fast Fourier transform is the advancement of the Fourier transform. The n by n transformation matrix is broken down into several levels. The reduced transformation matrix will be of order $n\text{-log}(n)$. The $n\text{-log}(n)$ matrix comprises identity, diagonal, functional matrixes, making them very easy to calculate because of the matrix properties. Thus it reduces the calculation time.

The FFT will convert the signal into a complex equation, which in turn, the data will be the known form of sin and cosine. After the FT procedure, the signal's energy, defined as, is retained. The power spectral density (PSD) is the square of the absolute value of the FT amplitude and reflects the power concentrated at different frequencies in the signal.

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int f(t)e^{-j\omega t} dt \qquad |f(\omega)|^2$$

The fast Fourier transform that transform the signal of time series into frequency and energy scale. The above equation is used to get the frequency and power density.

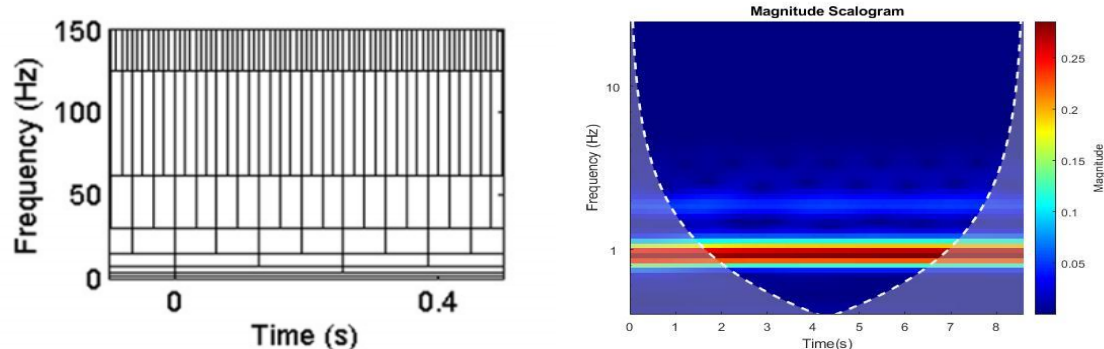


6-1 Fast Fourier transform for the angle of attack 15°

The FFT presents the frequency data in the signal with the power density. It does not feature the temporal data of the signal. Frequency occurrence with the temporal data is not exhibited in the power spectral plot. The FFT calculates the overall amplitude/phase at a frequency by integrating over the entire signal period, but it does not show how this amplitude/phase may change over time (Subash Chandran KS, 2016).

6.3. *Continuous Wavelet Transform*

The drawback of FFT is considered in the continuous wavelet transform that is the temporal data along with the frequency domain data are inferred with the energy density in the colour coding. The low frequency data are considered for the overall data set and as the frequency increase the time considered will be less to get the accurate result. The time resolution will be high based on the vertical position in the scalogram.



6-2 Continuous wavelet transform for the angle of attack for 15°

The fig shows the CWT plot or scalogram, the drawback is that the data obtain will be in the form of colour strip or a Gabor window. The size of the window depends on the vertical position in the mother wavelet equation.

6. 4. *Empirical Mode Decomposition*

Any complicated data collection can be reduced into a finite and often small number of "intrinsic mode functions" that permit well-behaved Hilbert transforms using the "empirical mode decomposition (EMD)" method. This decomposition strategy is adaptive and, as a function, extremely effective. Because its decomposition is based on the data's local characteristic time scale, it can be used for nonlinear and non-stationary data.

The decomposition is the method of extraction of data associated with energy at the different intrinsic time frames, which are the vital parameter of the system. They have well-behaved Hilbert transforms when expressed in IMFs, from which the instantaneous frequencies can be computed. As a result, it can be located at any event on both the time and frequency axes.

a) Intrinsic Mode Function (IMF):

An intrinsic mode function must satisfy two essential conditions:

- (1) The zero crossings and the extrema numbers in the signal should be the same or differ by one.
- (2) The envelope created by the local maxima and minima should have the zero-mean line at any point.

The first criterion is self-evident; it resembles the narrow band criteria for a stable Gaussian process. The second condition is a concept; it converts a global requirement to a local requirement to prevent undesirable fluctuation in the instantaneous frequency caused by asymmetric wave shapes. The ideal criterion would be for the data's local mean to be zero. The 'local mean' for non-stationary data requires the computation of a 'local time scale,' which is impossible to specify. Instead, force the local symmetry using the local mean of the envelopes generated by the local maxima and minima as a surrogate. This is a necessary estimate to avoid defining a local averaging time scale.

b) Instantaneous Frequency and Hilbert transform:

The acceptance of the concept of an instantaneous frequency involves two functions. The first is the result of the Fourier spectral analysis's extensive effect. The frequency of a sin or cosine function spanning the entire data length with constant amplitude is specified in traditional Fourier analysis. The instantaneous frequencies must also be related to either a sin or a cosine function to extend this notion. To define the local frequency value, needed at least one complete oscillation of a sine or cosine wave. Nothing less than a full-wave will suffice, according to this logic. For non-stationary data, such a concept would be absurd.

The main factor stems from the fact that the instantaneous frequency is defined in a non-unique way. Nonetheless, since the development of the Hilbert transforms as a technique of making data analytical, this problem is no longer a significant issue. The Hilbert Transform, $Y(t)$, of any arbitrary time series $X(t)$, may always be found as

$$Y(t) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{X(t')}{t - t'} dt'$$

The Cauchy principal value is denoted by the letter P. With this concept, $X(t)$ and $Y(t)$ form the complex conjugate pair, and $Z(t)$ can be used as an analytic signal. Where $Z(t)$ is

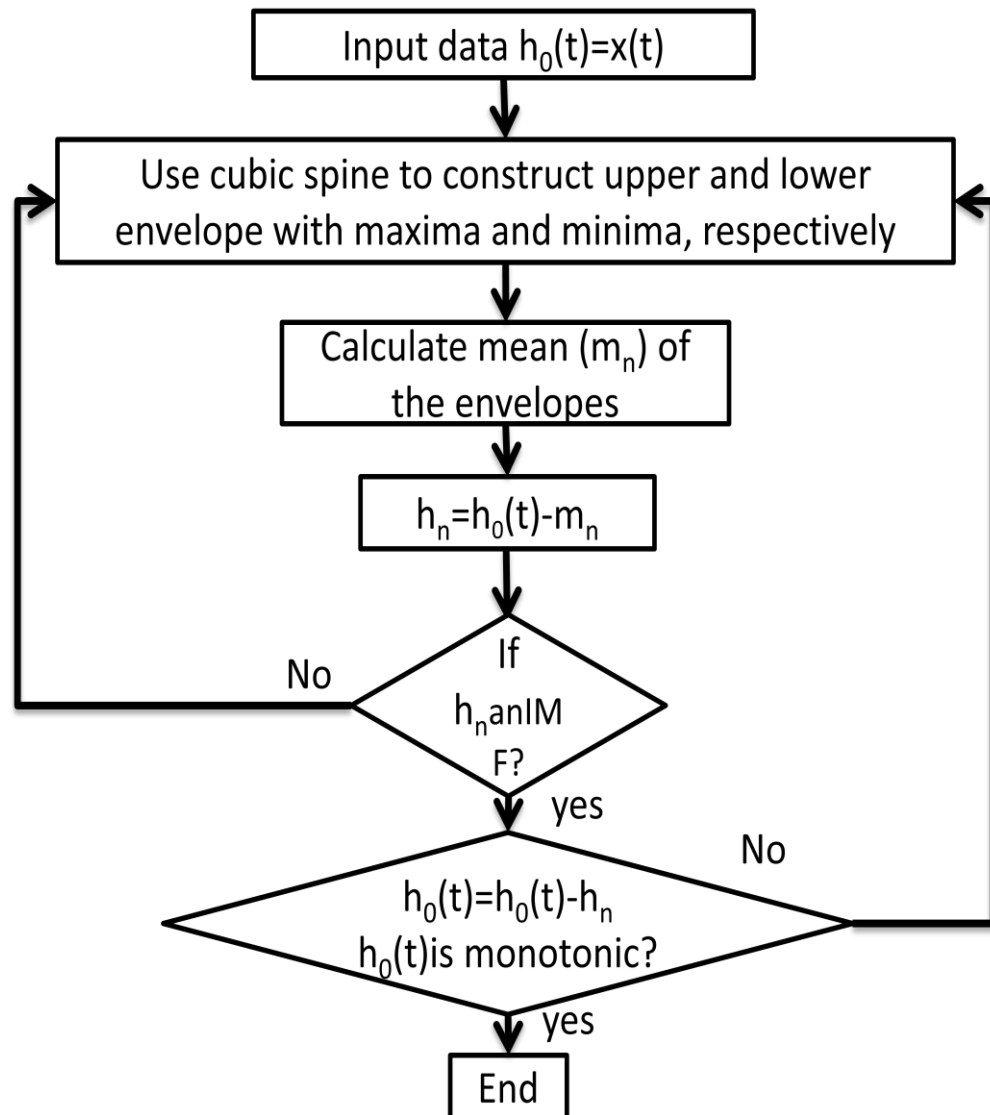
$$Z(t) = X(t) + iY(t) = a(t)e^{ti\theta}$$

In which,

$$a(t) = [X^2(t) + Y^2(t)]^{1/2}, \quad \theta(t) = \arctan\left(\frac{Y(t)}{X(t)}\right)$$

The Hilbert transform is a unique method of defining the imaginary portion, resulting in an analytic function. In the above equation, $a(t)$ represents the power density in the analytic signal, and the $\theta(t)$ will give information about the position of that analytical signal (Tzu-Hsun Hung, 2012).

c) Algorithm of EMD;



The above flow diagram represents the algorithm of the EMD technique. The input signal $x(t)$ time-variant data is considered as $h_0(t)=x(t)$. The upper and lower envelope is created over the signal by connecting the minima and maxima points. The piecewise cubic hermit interpolation polynomial connects the points of maxima and minima. The envelope consists of all the signal data points in it (Tzu-Hsun Hung, 2012).

The mean of the envelope m_n is calculated at each instant. The mean (m_n) is deducted in the given input signal in the algorithm. That is denoted as h_n , where

$$h_n = h_0(t) - m_n$$

If h_n satisfies the condition of IMF, the data is stored and plotted, and the Hilbert Transform is performed on the IMF. Else h_n is looped with-in from creating an envelope to the deduct of mean until the IMF condition is satisfied. This loop makes the signal from varying mean to zero mean, preventing undesirable changes in the instantaneous frequency.

The IMF h_n that is recorded will be deducted from the initial data, and it will be assigned as $h_0(t)$

$$h_0(t) = h_0(t) - h_n$$

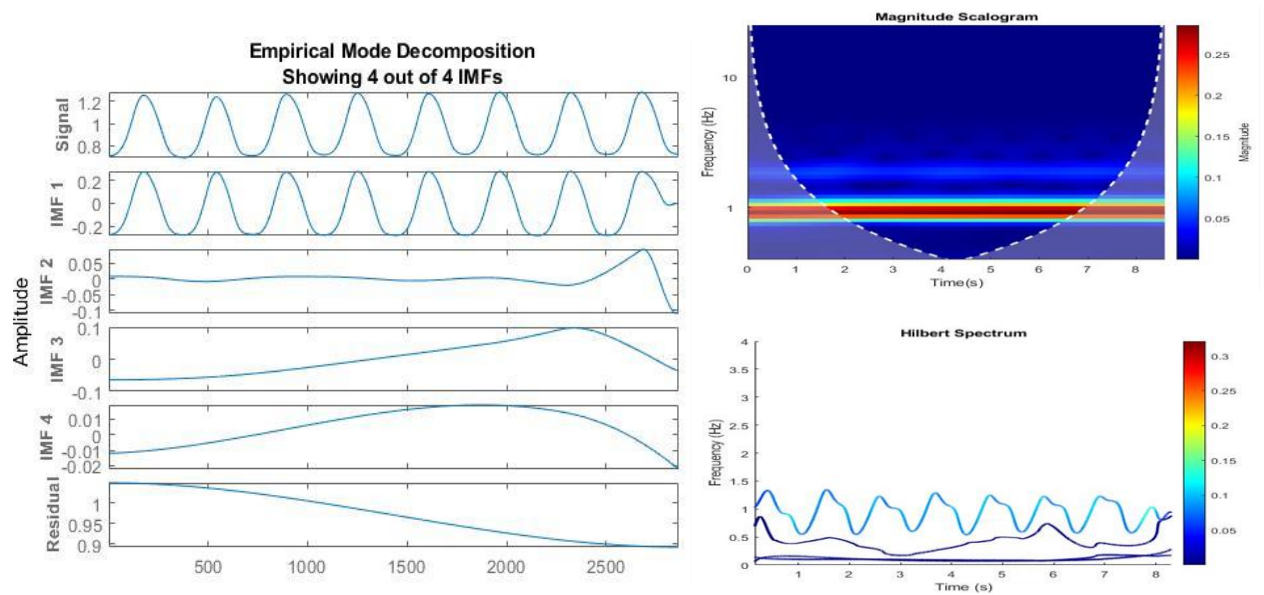
On further simplification of above equation

$$h_0 = \sum_{p=1}^n m_p$$

A monotonic signal is a condition where the signal will either ascend or descend. If the new h_0 from the above equation satisfies the monotonic requirement, the loop end ends. Otherwise, the h_0 cumulative mean of the loop is treated as the new input signal and run through the algorithm till the second loop condition satisfies. At the last loop the final h_0 is assigned as residual.

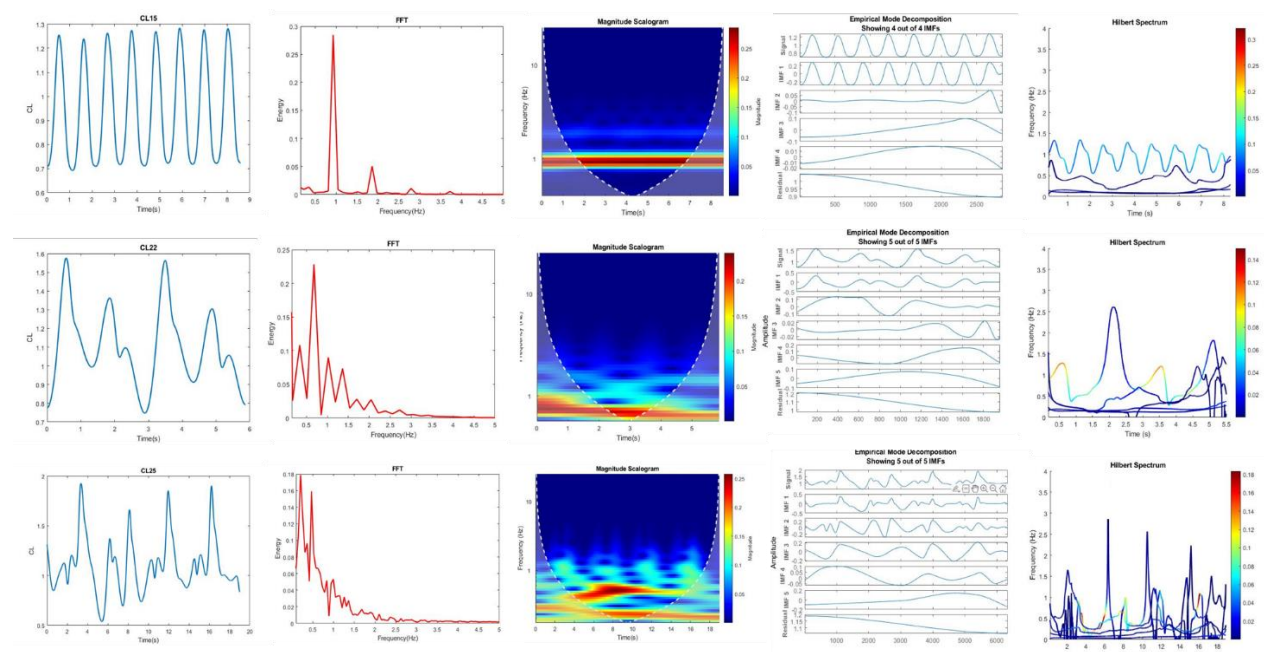
This EMD algorithm decomposes the signal from the higher frequency to lower with the zero mean, making the signal study very easy after performing the Hilbert Transform. All the IMF's Hilbert transform is stacked together, which gives the instantaneous frequencies with its power density that is available at every particular point.

EMD provides the time domain data and frequency domain data with higher accuracy even for non-linear and non-stationary data. In contrast, the FFT will exhibit the frequency data, and the details of the temporal data are not available. It does not present how the frequency of this event may alter over time.



6-3 Comparison of CWT with EMD and HHT

The above fig shows the comparison of the HHT and CWT, HHT gives the instantaneous frequency and energy with respect to time, where CWT provides the frequency in the form of colour band and the resolution will be high for high frequency component.



6-4 time series analysis for different AOA 15° (top row), 22° (center row), 25° (bottom row), CL data set (1st column), FFT (2nd column), CWT (3rd column), EMD (4th column), HHT (5th column).

The various angle of attack are shown from periodic to chaotic data which are 10°, 22° and 25°. It can be seen that instantaneous frequency is clearly available in the HHT compared to other techniques.

7. Conclusion

Analysis tools help us in understanding the contribution of flow field in lift generation and vortex shedding and to observe the transition from periodicity to chaos.

For smaller angles of attack, higher recurrence rate and determinism in the Recurrence Plot was found. As the AoA increases, the recurrence rate and determinism drastically reduce indicating a decrease in the lift and an increase in the drag.

Recurrence is observed when a data set repeats itself over time. Simulation is a long process requiring lots of resources, recurrence plot can be used to some degree to narrow down the AoA or at least eliminate some the AoAs, thus reducing the requirement of time and resources.

Time series analysis is a way to separate the low and high frequencies in the signal and its energy density to infer the effect of turbulence and coherent behaviour of the vortices based on the Strouhal Number which is directly proportional to frequency for the better understanding.

An increasing trend from an AoA of 0° to 20° was observed and at an AoA of 23° and 24° a sudden increase in Cl value was observed. The physical phenomenon behind this increase will be the scope of a future study.

The results obtained could be used for various applications such as UAV, MVA etc.

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9. MATLAB Syntax

Recurrence Quantification Analysis

```
tau = mdDelay (X,'maxLag',2000,'plotttype','all')
[fnnperc,emTimes] = mdFnn(X, first minima)
phasespace(X)
crp(X)
crqa(X)
```

Fast Fourier Transform

```
X=data;
dt = 0.003;
L=length(X);
t=0: dt (L-1)* dt;
fs = 1/dt;

Y=fft (X);
P2 = abs (Y/L);
P1 = P2 (1: L/2+1);
P1 (2: end-1) = 2*P1 (2: end-1)
f = fs*(0: (L/2))/L;
plot(f, P1);
```

Continuous Wavelet transform

```
Cwt (X, fs)
```

Empirical Mode Decomposition

```
[imf, residual, info] = emd (X, 'Interpolation', 'pchip');

emd(X, 'Interpolation' , 'pchip', 'MaxNumIMF' , 10)
```

Hilbert Hung Transform

```
[hs, f, t] = hht(imf , fs);

mesh(seconds(t), f, hs, 'EdgeColor' , 'none', 'FaceColor', 'interp')

xlabel ('time (s)')
ylabel ('Frequency (Hz)')
zlabel ('Instantaneous Energy')
hht(IMF, fs)
```